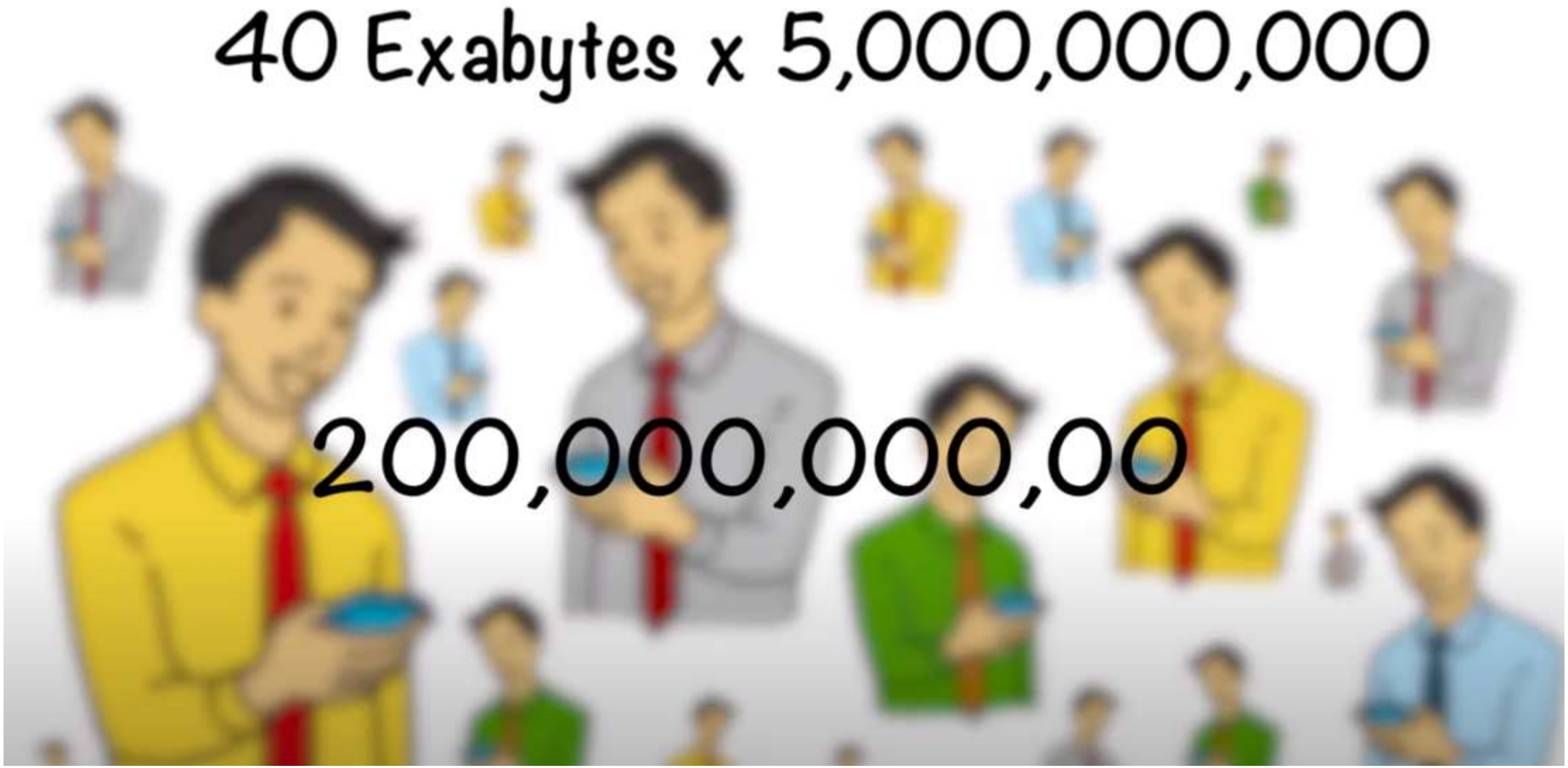
# Granularity of Data Items

The size of data items chosen as the unit of protection by a concurrency protocol. Granularity can range from small to large data items e.g. the entire database, a file, a page (sometimes called an area or database space - a section of physical disk in which relations are stored), a record or a field value of a record. This size has a very significant effect in the designing of concurrency algorithms.

https://www.youtube.com/watch?v=q_-fhQXOQbg

# Big Data


https://www.youtube.com/watch?v=bAyrObl7TYE




https://www.ionos.com/digitalguide/websites/web-development/what-is-an-exabyte/

| Decimal prefix (SI) | In byte + conversion factor |
|---|---|
| Kilobyte (KB) | 1,000 ($10^3$) |
| Megabyte (MB) | 1,000,000 ($10^6$) |
| Gigabyte (GB) | 1,000,000,000 ($10^9$) |
| Terabyte (TB) | 1,000,000,000,000 ($10^{12}$) |
| Petabyte (PB) | 1,000,000,000,000,000 ($10^{15}$) |
| Exabyte (EB) | 1,000,000,000,000,000,000 ($10^{18}$) |
| Zettabyte (ZB) | 1,000,000,000,000,000,000,000 ($10^{21}$) |
| Yottabyte (YB) | 1,000,000,000,000,000,000,000,000 ($10^{24}$) |

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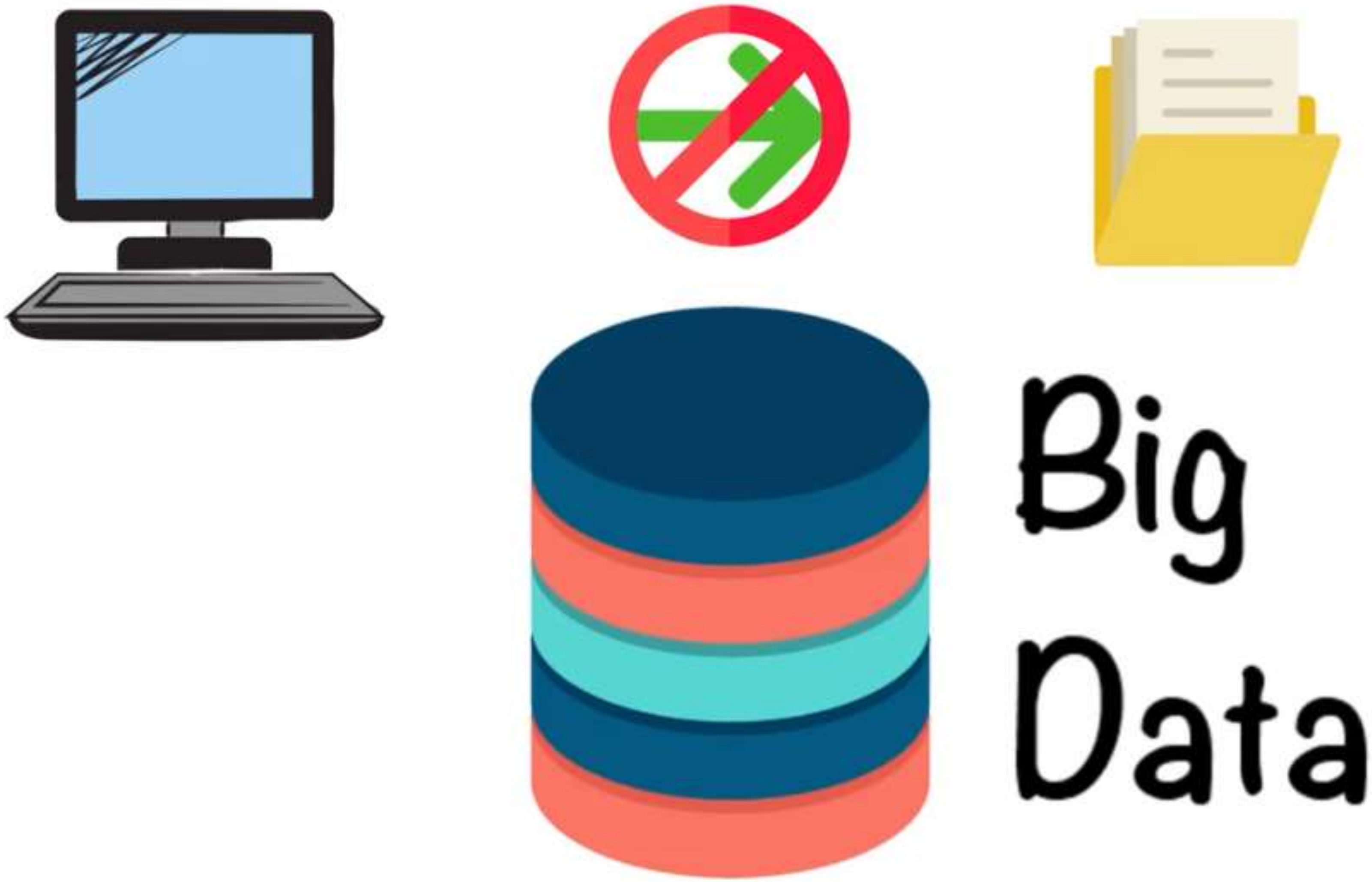


https://www.ionos.com/digitalguide/websites/web-development/what-is-an-exabyte/


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# This is possible with the concept of


Volume

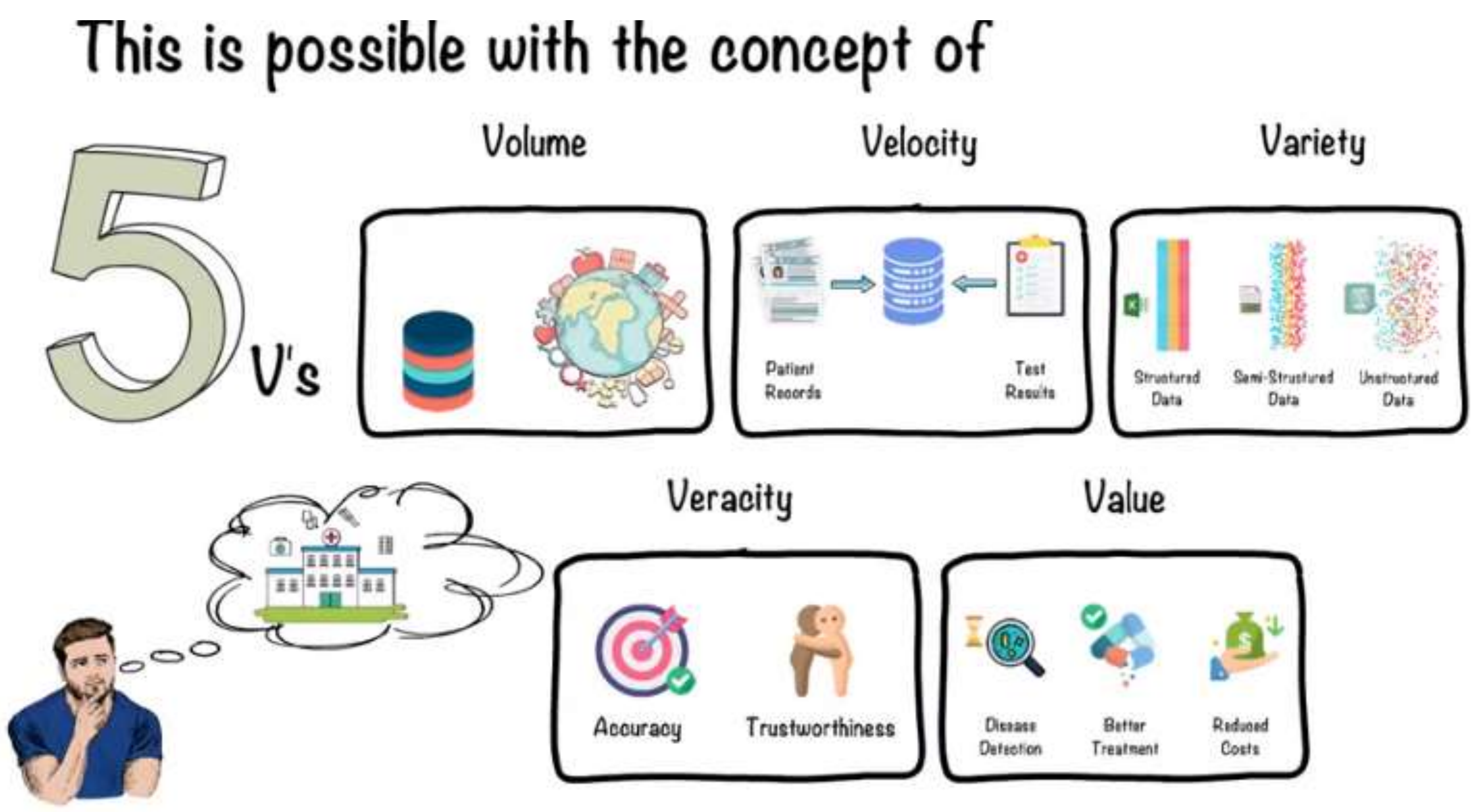
Velocity

Variety

Veracity

Value

https://www.youtube.com/watch?v=bAyrObl7TYE




https://www.youtube.com/watch?v=bAyrObl7TYE

# Emerging Trends and Applications

# Emerging Trends and Applications

1. Artificial Intelligence(AI)
2. Big data
3. Internet Of Things(IoT)
4. Cloud Computing
5. Grid Computing
6. Blockchains

https://www.youtube.com/watch?v=fMVpOPiVmCU

# Artificial Intelligence

- The intelligent digital personal assistants like Siri, GoogleNow, Cortana, and Alexa are all powered by AI.
- Artificial Intelligence endeavours to simulate the natural intelligence of human beings into machines, thus making them behave intelligently.
- An intelligent machine is supposed to imitate some of the cognitive functions of humans like learning, decision making, and problem-solving.
- In order to make machines perform tasks with minimum human intervention, they are programmed to create a knowledge base and make decisions based on it.

https://www.youtube.com/watch?v=fMVpOPiVmCU

# Big data

- With technology making an inroad into almost every sphere of our lives, data is being produced at a colossal rate.
- Today, there are over a billion Internet users, and a majority of the world's web traffic is coming from smartphones.
- Around 2.5 quintillion bytes of data are created each day, and the pace is increasing with the continuous evolution of the Internet of Things(IoT).
- This results in the generation of data sets of enormous volume and complexity called *Big Data*.

https://www.youtube.com/watch?v=fMVpOPiVmCU

# Internet of Things

- The term computer network that we commonly use is the network of computers. Such a network consists of a laptop, desktop, server, or a portable device like a tablet, smartphone, smartwatch, etc., connected through wired or wireless. We can communicate between these devices using the Internet or LAN. The 'Internet of Things' is a network of devices that have embedded hardware and software to communicate (connect and exchange data) with other devices on the same network.
- These devices are used in isolation from each other, with a maximum human intervention needed for operational directions and input data.
- IoT tends to bring together these devices to work in collaboration and assist each other in creating an intelligent network of things.

https://www.youtube.com/watch?v=fMVpOPiVmCU

# Cloud Computing

- Cloud computing is an emerging trend in the field of information technology, where computer-based services are delivered over the Internet or the cloud, and it is accessible to the user from anywhere using any device.
- The services comprise software, hardware (servers), databases, storage, etc.
- These resources are provided by companies called cloud service providers and usually charge on a pay-per-use basis, like the way we pay for electricity usage.
- A user can run a bigger application or process a large amount of data without having the required storage or processing power on their personal computer as long as they are connected to the Internet.

**Beside other numerous features, cloud computing offers cost effective on demand resources.**

https://www.youtube.com/watch?v=fMVpOPiVmCU